SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: **Artificial Intelligence** **COURSE CODE: CSA17**

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively.*(BL3)*

Assessment Tool Weightage: 50%

QUESTIONS: 2

Duration : 1 Hour
Total Marks:30

Annexure A

<u>Constraint-Based Problem Solving</u>

Code of Conduct:

I G. Yeshwanth Ram. 192311397 *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

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Co1 Assessment 02

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1, Map coloring using Backtracking Search (Constraint Satisfaction Problem)

Given:

· Regions : R_1, R_2, R_3, R_4

Colors : {Red(R), Green(G), Blue(B)}

Constraints:

1. Adjacent regions cannot have the same color:

· $R1 \Leftrightarrow R2$

· $R2 \Leftrightarrow R3$

· $R3 \Leftrightarrow R4$

· $R4 \Leftrightarrow R1$

2. R1 = Red

3. $R_3 \neq$ Blue

4. Every region gets exactly one color

Variable Order

$R_1 \Leftrightarrow R2 \Leftrightarrow R_3 \Leftrightarrow R4$

Assign ~~for~~ R_1

· Constraint : R_1 must be Red

· Assign R_1 = Red

Current Assignment :

R_1 = Red

R_2 = ?

R_3 = ?

R4 ?

Assign R_2

Possible colors = {Red, Green, Blue}

. Try Red :

- Adjacent to R1 (Red)

- Conflict x

- Back track

· Try Green

· Different from R1

· Valid ✓

Assignment :

R1 = Red , R_2 = Green

R_3 = ? , R_4 = ?

Step 3: Assign R_3

Possible colors: {Red, Green, Blue}

Constraint:

- Cannot be Blue
- Adjacent to R_2 (Green)

Try Green

- Same as R_2
- Conflict ✗

· Try Blue

Not allowed

Constraint violated

· Try Red:

- Different from Green
- Allowed

Assignment:

R_1 = Red

R_2 = Green

R_3 = Red

R_4 = ?

Step 4: Assign R_4

Possible colors: {Red, Green, Blue}

Constraints:

- Adjacent to R3 (Red)
- Adjacent to R1 (Red)
- Try Red
- Same as R1 and B3
- Conflict ✗
- Try Green
- Different from R1 and R_3
- Valid ✓

No Need to test Blue since a valid color is already found.

Final Assignment:

R_1 = Red

R_2 = Green

R_3 = Red

R_4 = Green

BFS Search with Manhattan Distance:

Since the question does not provide the 5x5 grid or obstacle locations, the exact BFS expansion and optimal path cannot be uniquely determined. Below is a standard worked example

Grid:

```
S . . X .
. X . X .
. X . . .
. . X X .
. . . . G
```

where

- S = Start (0,0)
- G = Goal (4,4)
- X = Obstacle
- '.' = free cell.

Manhattan distance formula.

$$h(n) = |x_{goal} - x| + |y_{goal} - y|$$

Example:

For start (0,0)

$$h = |4-0| + |4-0| = 8$$

BFS execution:

Initial state

Queue = [(0,0)]

Visited = {(0,0)}

Expand (0,0)

Possible moves:

- Right (0,1) ✓
- Down (1,0) ✓

Queue:

[(0,1), (1,0)]

Expand (0,1)

New Node

(0,2)

Queue

[(1,0), (0,2)]

Expand (1,0)

New Node

(2,0)

Queue

[(0,2), (2,0)]

Expand (0,2)

New Node

(1,2)

Queue

[(2,0), (1,2)]

Expand (2,0)

New Node

(3,0)

Queue

[(1,2), (3,0)]

Expand (1,2)

New Node

(2,2)

∴ The Search proceeds level by level until (4,4) is reached

Queue:

[(3,0), (2,2)]

Total Cost

Each move cost = 1

Number of moves = 8

Total path cost = 8

Manhattan Distance Values:

Node	h(n)
(0,0)	8
(1,0)	7
(2,0)	6
(3,0)	5
(4,0)	4
(4,1)	3
(4,2)	2
(4,3)	1
(4,4)	0